

Unit D Review
Multiple Choice Bank — 75 Questions

Question Set

MC 1. According to the lesson material, which condition is NOT essential for chemical equilibrium to be established?

- A. The system must be closed
- B. The temperature must remain above 400°C
- C. Observable properties must remain constant
- D. Forward and reverse reaction rates must be equal

Correct answer: B

Rationale: The lesson states three essential criteria for equilibrium: closed system, constant observable properties, and equal forward/reverse rates. Temperature of 400°C was specific to the ammonia example, not a general requirement.

MC 2. In the ammonia synthesis equilibrium $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$, what happens to the forward reaction rate as equilibrium is approached?

- A. It increases exponentially
- B. It remains constant
- C. It decreases due to decreasing reactant concentrations
- D. It becomes zero at equilibrium

Correct answer: C

Rationale: The lesson explicitly states that as the forward reaction proceeds, $[\text{N}_2]$ and $[\text{H}_2]$ decrease, reducing collision frequency and thus the forward reaction rate.

MC 3. The lesson compares chemical equilibrium to a crowded dance floor. What aspect of equilibrium does this analogy illustrate?

- A. Static balance with no molecular movement
- B. Complete conversion of reactants to products
- C. Temperature dependence of equilibrium
- D. Dynamic balance with constant overall numbers despite individual movement

Correct answer: D

Rationale: The dance floor analogy shows people entering and leaving at the same rate, maintaining constant numbers while individuals move - illustrating dynamic equilibrium where concentrations stay constant despite ongoing reactions.

MC 4. According to Le Chatelier's principle discussed in the lesson, how does a system respond to external changes?

- A. By permanently stopping all reactions
- B. By shifting to counteract the imposed change
- C. By increasing temperature only
- D. By converting all reactants to products

Correct answer: B

Rationale: The lesson states that Le Chatelier's principle operates through rate changes that temporarily unbalance equilibrium, driving net reaction to counteract imposed changes.

MC 5. What is the bond energy of the N_2 triple bond mentioned in the ammonia synthesis discussion?

- A. 391 kJ/mol

- B. 436 kJ/mol
- C. 945 kJ/mol
- D. 150 kJ/mol

Correct answer: C

Rationale: The lesson specifically states that the N₂ triple bond has a bond energy of 945 kJ/mol.

MC 6. In the Haber-Bosch process described, what pressure range is used for industrial ammonia synthesis?

- A. 150-200 atm
- B. 400-500 atm
- C. 1-10 atm
- D. 50-75 atm

Correct answer: A

Rationale: The lesson states that the Haber-Bosch process operates under carefully controlled pressure of 150-200 atm.

MC 7. According to the Brønsted-Lowry definition mentioned in the lesson, what characterizes an acid-base reaction?

- A. Electron transfer between molecules
- B. Proton transfer between molecules
- C. Formation of ionic bonds
- D. Breaking of covalent bonds only

Correct answer: B

Rationale: The lesson explicitly states that Brønsted-Lowry acid-base reactions involve proton transfer between molecules.

MC 8. What pH range must blood maintain for proper physiological function according to the lesson?

- A. 6.8-7.2
- B. 7.0-7.4
- C. 7.35-7.45
- D. 7.5-8.0

Correct answer: C

Rationale: The lesson specifically states that blood pH must remain 7.35-7.45 for proper physiological function.

MC 9. How do buffer systems maintain constant pH according to the lesson material?

- A. Through temperature changes only
- B. By precipitating excess ions
- C. Through weak acid-conjugate base equilibria that neutralize added acids or bases
- D. By converting all acids to neutral salts

Correct answer: C

Rationale: The lesson states that buffer systems maintain constant pH through weak acid-conjugate base equilibria that chemically neutralize added acids or bases.

MC 10. What determines whether equilibrium favors reactants or products according to the molecular-level discussion?

- A. Only the temperature of the system
- B. The overall energy balance of bond breaking and forming
- C. The color of the products
- D. The volume of the container alone

Correct answer: B

Rationale: The lesson explains that the overall energy balance between bond breaking and forming determines whether equilibrium favors reactants or products.

MC 11. According to the lesson, what happens to equilibrium constant (K_c) values when concentration changes occur?

- A. K_c increases proportionally
- B. K_c decreases proportionally
- C. K_c remains unchanged
- D. K_c becomes zero

Correct answer: C

Rationale: The lesson emphasizes that K_c values depend only on temperature while equilibrium position can be shifted by concentration changes without changing K_c .

MC 12. What temperature range is used in the Haber-Bosch process for ammonia synthesis?

- A. 100-200°C
- B. 200-300°C
- C. 300-400°C
- D. 400-500°C

Correct answer: D

Rationale: The lesson states that the Haber-Bosch process operates at temperatures of 400-500°C.

MC 13. In the ammonia equilibrium system, the reverse rate increases proportionally to what?

- A. $[N_2][H_2]^3$
- B. $[NH_3]^2$
- C. $[NH_3]$
- D. $[N_2]^2[H_2]$

Correct answer: B

Rationale: The lesson specifically states that the reverse rate increases proportionally to $[NH_3]^2$.

MC 14. What is the bond energy of each N-H bond in ammonia according to the lesson?

- A. 945 kJ/mol
- B. 436 kJ/mol
- C. 391 kJ/mol
- D. 150 kJ/mol

Correct answer: C

Rationale: The lesson states that each N-H bond in ammonia has a bond energy of 391 kJ/mol.

MC 15. According to the lesson, equilibrium constants quantify what aspect of a reaction?

- A. Reaction speed only
- B. Thermodynamic favorability
- C. Color intensity
- D. Molecular size

Correct answer: B

Rationale: The lesson states that equilibrium constants quantify thermodynamic favorability and depend only on temperature.

MC 16. What type of distribution affects equilibrium through temperature according to the lesson?

- A. Maxwell-Boltzmann distribution
- B. Normal distribution

- C. Poisson distribution
- D. Uniform distribution

Correct answer: A

Rationale: The lesson specifically mentions that temperature affects equilibrium through the Maxwell-Boltzmann distribution of molecular kinetic energies.

MC 17. In the ammonia synthesis, what initially causes the forward rate to be much greater than the reverse rate?

- A. High temperature only
- B. High $[N_2]$ and $[H_2]$ while $[NH_3]$ is zero
- C. Presence of catalyst
- D. Low pressure conditions

Correct answer: B

Rationale: The lesson states that initially, the forward reaction dominates because $[N_2]$ and $[H_2]$ are high while $[NH_3]$ is zero.

MC 18. What does the lesson identify as the impact of not understanding equilibrium on fertilizer production?

- A. Increased color intensity
- B. Faster reaction rates only
- C. Efficient large-scale production would be impossible
- D. Lower temperatures would be needed

Correct answer: C

Rationale: The lesson states that without equilibrium understanding, efficient large-scale fertilizer production would be impossible, directly impacting global food security.

MC 19. According to the lesson, buffer effectiveness depends on having what structural components?

- A. Only strong acids
- B. Only strong bases
- C. Both weak acid and conjugate base structures
- D. Neutral salts only

Correct answer: C

Rationale: The lesson states that buffer effectiveness depends on having both weak acid and conjugate base structures available for neutralization reactions.

MC 20. What determines acid-base behavior according to the structure-function relationship in the lesson?

- A. Molecular structure through electronegativity patterns
- B. Temperature alone
- C. Container volume
- D. Reaction time

Correct answer: A

Rationale: The lesson states that molecular structure determines acid-base behavior, with electronegativity patterns creating polar bonds for proton donation/acceptance.

MC 21. How long does the ammonia synthesis typically take to reach equilibrium according to the lesson?

- A. Several seconds
- B. Several minutes to hours
- C. Several days
- D. Several weeks

Correct answer: B

Rationale: The lesson states that the ammonia synthesis system typically reaches equilibrium after several minutes to hours depending on temperature and catalyst presence.

MC 22. What principle explains that systems naturally evolve toward states of minimum free energy?

- A. Le Chatelier's principle
- B. Conservation of mass
- C. Thermodynamic principle
- D. Kinetic molecular theory

Correct answer: C

Rationale: The lesson states that equilibrium reflects the thermodynamic principle that systems naturally evolve toward states of minimum free energy while maximizing entropy.

MC 23. In the dance floor analogy, what represents the equilibrium state?

- A. Everyone stops dancing
- B. The number of dancers stays constant while people enter and leave at the same rate
- C. Only entering is allowed
- D. The dance floor becomes empty

Correct answer: B

Rationale: The lesson uses the dance floor analogy where people enter and leave at exactly the same rate, keeping the number of dancers constant, representing equilibrium.

MC 24. What happens to reactant concentrations as the ammonia synthesis proceeds toward equilibrium?

- A. They increase linearly
- B. They remain constant
- C. They decrease exponentially
- D. They fluctuate randomly

Correct answer: C

Rationale: The lesson states that as time progresses toward equilibrium, reactant concentrations decrease exponentially.

MC 25. What is the bond energy of H_2 mentioned in the ammonia synthesis discussion?

- A. 391 kJ/mol
- B. 436 kJ/mol
- C. 945 kJ/mol
- D. 200 kJ/mol

Correct answer: B

Rationale: The lesson specifically states that the H_2 bond has an energy of 436 kJ/mol.

MC 26. According to the lesson, what determines the direction of acid-base reactions in the Brønsted-Lowry model?

- A. Temperature only
- B. Relative proton affinities and equilibrium constants
- C. Molecular weight
- D. Color changes

Correct answer: B

Rationale: The lesson states that reaction direction is determined by relative proton affinities and equilibrium constants in Brønsted-Lowry acid-base reactions.

MC 27. How does product concentration increase as equilibrium is approached according to the lesson?

- A. Exponentially

- B. Linearly
- C. Logarithmically
- D. It doesn't increase

Correct answer: C

Rationale: The lesson states that product concentration increases logarithmically as the system approaches equilibrium.

MC 28. What must be overcome for N_2 and H_2 to form ammonia according to the energy discussion?

- A. Only entropy barriers
- B. Bond energies in N_2 triple bond and H_2 bond
- C. Temperature gradients
- D. Pressure differences only

Correct answer: B

Rationale: The lesson states that bond energies in the N_2 triple bond (945 kJ/mol) and H_2 bond (436 kJ/mol) must be overcome to form N-H bonds.

MC 29. According to the lesson, what creates dynamic equilibrium in chemical systems?

- A. Complete cessation of all reactions
- B. Forward reactions only
- C. Both forward and reverse reactions occurring at the same rate
- D. Temperature changes alone

Correct answer: C

Rationale: The lesson states that chemical equilibrium occurs when molecules undergo both forward and reverse reactions at the same rate, creating dynamic balance.

MC 30. What mathematical relationships does the lesson mention for acid-base equilibrium systems?

- A. Only pH calculations
- B. Relationships between pH, pOH, K_a , K_b , and K_w
- C. Temperature equations only
- D. Pressure-volume relationships

Correct answer: B

Rationale: The lesson specifically mentions mathematical relationships between pH, pOH, K_a , K_b , and K_w for quantitative prediction of acid-base systems.

MC 31. What does the lesson identify as enabling biological function and industrial process control?

- A. High temperatures only
- B. Strong acids
- C. Buffer systems
- D. Catalysts alone

Correct answer: C

Rationale: The lesson states that buffer systems maintain constant pH, enabling biological function and industrial process control.

MC 32. In equilibrium establishment, what reduces collision frequency according to the lesson?

- A. Increasing product concentration
- B. Decreasing reactant concentrations
- C. Temperature increase
- D. Adding catalyst

Correct answer: B

Rationale: The lesson states that as $[N_2]$ and $[H_2]$ decrease, this reduces collision frequency and thus the forward reaction rate.

MC 33. What does the lesson say about the forward rate's proportionality in ammonia synthesis?

- A. Proportional to $[NH_3]^2$
- B. Proportional to $[N_2][H_2]^3$
- C. Proportional to temperature only
- D. Inversely proportional to pressure

Correct answer: B

Rationale: The lesson states that the forward rate decreases proportionally to $[N_2][H_2]^3$.

MC 34. According to the lesson, what characterizes the concentration-time curve at equilibrium?

- A. It continues rising indefinitely
- B. It drops to zero
- C. It levels off at constant values
- D. It oscillates regularly

Correct answer: C

Rationale: The lesson describes the characteristic concentration-time curve that levels off at constant values when equilibrium is reached.

MC 35. What does the lesson identify as the basis for pH control in buffer systems?

- A. Strong acid-strong base reactions
- B. Weak acid-conjugate base equilibria
- C. Temperature changes
- D. Pressure variations

Correct answer: B

Rationale: The lesson states that buffer systems utilize weak acid-conjugate base equilibria for pH control.

MC 36. According to the lesson, what happens to molecules at equilibrium?

- A. All reactions stop completely
- B. Only forward reactions continue
- C. Individual molecules continue to react while concentrations remain constant
- D. Molecules become permanently fixed

Correct answer: C

Rationale: The lesson emphasizes that at equilibrium, individual molecules continue to react (dynamic) while concentrations remain constant.

MC 37. What does increasing temperature do to molecules according to the Maxwell-Boltzmann discussion?

- A. Decreases all molecular energies
- B. Increases the fraction of molecules with sufficient energy to overcome activation barriers
- C. Stops all molecular motion
- D. Only affects product molecules

Correct answer: B

Rationale: The lesson states that higher temperatures increase the fraction of molecules possessing sufficient energy to overcome activation barriers.

MC 38. What prevents disruption of equilibrium according to the closed system requirement?

- A. Temperature control only
- B. Preventing reactants/products from escaping or new substances from entering

- C. Adding more catalyst
- D. Changing the container color

Correct answer: B

Rationale: The lesson states that a closed system prevents reactants or products from escaping or new substances from entering, which would disrupt the balance.

MC 39. According to the lesson, what aspect of equilibrium do chemical engineers optimize in the Haber-Bosch process?

- A. Container color
- B. Reaction conditions for maximum ammonia yield
- C. Only temperature
- D. Only pressure

Correct answer: B

Rationale: The lesson states that understanding equilibrium allows chemical engineers to optimize reaction conditions for maximum ammonia yield.

MC 40. What two factors does the lesson identify as determining equilibrium position through balance?

- A. Color and odor
- B. Volume and mass
- C. Enthalpy changes and entropy changes
- D. Time and distance

Correct answer: C

Rationale: The lesson states that the balance between enthalpy changes (bond breaking/forming) and entropy changes (molecular disorder) determines equilibrium position.

MC 41. According to the lesson, what happens when Le Chatelier's principle is applied to a system?

- A. Equilibrium is permanently destroyed
- B. Rate changes temporarily unbalance equilibrium until a new balance is established
- C. Temperature always increases
- D. All reactions stop

Correct answer: B

Rationale: The lesson states that Le Chatelier's principle operates through rate changes that temporarily unbalance equilibrium, driving net reaction until a new balance point is established.

MC 42. What does the lesson indicate about equilibrium constants and temperature?

- A. They are independent of temperature
- B. They depend only on temperature
- C. They change with concentration
- D. They change with pressure at constant temperature

Correct answer: B

Rationale: The lesson emphasizes that equilibrium constants depend only on temperature.

MC 43. In the ammonia synthesis, what provides molecules capable of decomposing in the reverse reaction?

- A. Decreasing N_2
- B. Decreasing H_2
- C. Increasing NH_3
- D. Adding catalyst

Correct answer: C

Rationale: The lesson states that increasing $[\text{NH}_3]$ provides more molecules capable of decomposing, increasing the reverse reaction rate.

MC 44. What observable properties does the lesson mention must remain constant at equilibrium?

- A. Only temperature
- B. Concentration, color, temperature, and pressure
- C. Only concentration
- D. Only pressure

Correct answer: B

Rationale: The lesson specifically lists concentration, color, temperature, and pressure as observable properties that must remain constant at equilibrium.

MC 45. According to the lesson, what enables quantitative prediction and optimization of acid-base systems?

- A. Color changes only
- B. Mathematical relationships between pH, pOH, K_a , K_b , and K_w
- C. Temperature measurements alone
- D. Pressure readings

Correct answer: B

Rationale: The lesson states that mathematical relationships between pH, pOH, K_a , K_b , and K_w enable quantitative prediction and optimization of acid-base equilibrium systems.

MC 46. What does the lesson say equilibrium constants reflect about structural differences?

- A. Color variations
- B. Stability differences
- C. Size differences only
- D. Shape differences only

Correct answer: B

Rationale: The lesson states that equilibrium constants reflect structural stability differences.

MC 47. According to the lesson, what is the initial reverse rate when ammonia synthesis begins?

- A. Maximum
- B. Equal to forward rate
- C. Zero
- D. Half of forward rate

Correct answer: C

Rationale: The lesson states that initially, $\text{Rate}_{\text{reverse}}$ is zero because no products exist.

MC 48. What does the lesson identify as the direct impact of equilibrium understanding on society?

- A. Better color in products
- B. Global food security through fertilizer production
- C. Faster transportation
- D. Improved entertainment

Correct answer: B

Rationale: The lesson states that equilibrium understanding enables fertilizer production, directly impacting global food security.

MC 49. According to the lesson, buffer systems respond to metabolic acid production through what mechanism?

- A. Temperature increase
- B. Chemical neutralization via weak acid-conjugate base equilibria

- C. Pressure changes
- D. Color changes

Correct answer: B

Rationale: The lesson indicates that buffer systems chemically neutralize added acids through weak acid-conjugate base equilibria.

MC 50. What does the lesson say about the relationship between molecular collision patterns and equilibrium?

- A. They are unrelated
- B. Collision patterns lead to equilibrium establishment
- C. Equilibrium prevents collisions
- D. Collisions only occur after equilibrium

Correct answer: B

Rationale: The lesson's cause-effect summary states that molecular collision patterns lead to equilibrium establishment.

MC 51. In the lesson's discussion of equilibrium, what creates polar bonds for proton donation/acceptance?

- A. Temperature gradients
- B. Pressure differences
- C. Electronegativity patterns
- D. Molecular size

Correct answer: C

Rationale: The lesson states that electronegativity patterns create polar bonds for proton donation/acceptance in acid-base behavior.

MC 52. According to the lesson, what maintains precise chemical equilibrium in living systems?

- A. Static conditions only
- B. Dynamic balance of opposing processes
- C. Complete reaction stoppage
- D. Random molecular motion

Correct answer: B

Rationale: The Big Biological Rule states that living systems maintain precise chemical equilibrium through dynamic balance of opposing processes.

MC 53. What does the lesson indicate about equilibrium position when pressure changes occur?

- A. It cannot shift
- B. K_c value changes
- C. Position can shift without changing K_c
- D. All equilibria are destroyed

Correct answer: C

Rationale: The lesson states that equilibrium position can be shifted by pressure changes without changing K_c .

MC 54. According to the lesson, what must be considered along with activation energy at the molecular level?

- A. Color intensity
- B. Collision frequency
- C. Container shape
- D. Atmospheric pressure

Correct answer: B

Rationale: The lesson states that equilibrium establishment involves collision frequency and activation energy considerations.

MC 55. What does the lesson say about the time when forward and reverse rates become equal?

- A. It marks the beginning of the reaction
- B. The system reaches equilibrium
- C. All reactions stop
- D. Temperature must be changed

Correct answer: B

Rationale: The lesson states that the system reaches equilibrium when $\text{Rate}_{\text{forward}} = \text{Rate}_{\text{reverse}}$.

MC 56. According to the lesson, cellular function depends on controlling equilibrium positions through what?

- A. Only temperature regulation
- B. Concentration changes, temperature regulation, and buffer systems
- C. Pressure changes only
- D. Color indicators

Correct answer: B

Rationale: The Big Biological Rule states that cellular function depends on controlling equilibrium positions through concentration changes, temperature regulation, and buffer systems.

MC 57. What does the lesson identify as necessary for neutralization reactions in buffers?

- A. Only strong acids
- B. Only strong bases
- C. Both weak acid and conjugate base structures
- D. Metal catalysts

Correct answer: C

Rationale: The lesson states that buffer effectiveness depends on having both weak acid and conjugate base structures available for neutralization reactions.

MC 58. According to the lesson, what type of balance characterizes chemical equilibrium?

- A. Static balance with no molecular activity
- B. Dynamic balance with ongoing molecular activity
- C. Temporary balance lasting seconds
- D. Unbalanced state

Correct answer: B

Rationale: The lesson emphasizes that equilibrium achieves dynamic balance with constant concentrations maintained by continuous molecular activity.

MC 59. What does the lesson say about the initial forward rate in ammonia synthesis?

- A. It is zero
- B. It is high due to maximum reactant concentrations
- C. It equals the reverse rate
- D. It is at minimum

Correct answer: B

Rationale: The lesson states that initially, $\text{Rate}_{\text{forward}}$ is high due to maximum reactant concentrations.

MC 60. According to the lesson, what allows optimization of industrial and biological systems?

- A. Random changes
- B. Mathematical quantification through equilibrium constants
- C. Color observation only
- D. Volume measurements alone

Correct answer: B

Rationale: The cause-effect summary indicates that mathematical quantification through equilibrium constants leads to optimization of industrial and biological systems.

MC 61. What does the lesson indicate about reactions when equilibrium concentrations level off?

- A. All reactions have stopped
- B. Forward and reverse reactions continue at equal rates
- C. Only forward reactions continue
- D. The system has failed

Correct answer: B

Rationale: The lesson emphasizes that when concentrations level off at equilibrium, forward and reverse reactions continue at equal rates (dynamic equilibrium).

MC 62. According to the lesson, what happens to entropy in equilibrium systems?

- A. It always decreases
- B. It becomes zero
- C. Systems naturally maximize it while minimizing free energy
- D. It is irrelevant to equilibrium

Correct answer: C

Rationale: The lesson states that systems naturally evolve toward states of minimum free energy while maximizing entropy.

MC 63. What does the lesson say about predictable responses in equilibrium systems?

- A. They are impossible to predict
- B. They result from external changes and can be quantified
- C. They only occur at high temperatures
- D. They require catalysts

Correct answer: B

Rationale: The cause-effect summary shows that equilibrium constants lead to predictable responses to external changes.

MC 64. According to the lesson, what resists pH disruption in biological systems?

- A. Strong acids only
- B. Temperature control
- C. Buffer systems
- D. Pressure regulation

Correct answer: C

Rationale: The Big Biological Rule mentions buffer systems that resist pH disruption in living systems.

MC 65. What does the lesson identify as the mechanism for precise pH control?

- A. Random acid-base reactions
- B. Buffer mechanisms using weak acid-conjugate base equilibria
- C. Strong acid-strong base titrations
- D. Temperature fluctuations

Correct answer: B

Rationale: The lesson indicates that precise pH control occurs through buffer mechanisms using weak acid-conjugate base equilibria.

MC 66. According to the lesson, what must be true about forward and reverse reaction rates at equilibrium?

- A. Forward rate exceeds reverse rate
- B. Reverse rate exceeds forward rate
- C. Both rates are exactly equal
- D. Both rates are zero

Correct answer: C

Rationale: The lesson states that one of the essential criteria for equilibrium is that forward and reverse reaction rates must be exactly equal.

MC 67. What does the lesson say about economically viable equilibrium in the Haber-Bosch process?

- A. It's impossible to achieve
- B. It requires balancing equilibrium concentrations with reasonable reaction rates
- C. It only depends on temperature
- D. It ignores equilibrium principles

Correct answer: B

Rationale: The lesson states that the process achieves economically viable equilibrium concentrations while maintaining reasonable reaction rates.

MC 68. According to the lesson, what happens to forward and reverse rates when rates intersect?

- A. The reaction stops
- B. Products convert completely to reactants
- C. Equilibrium occurs
- D. Temperature must increase

Correct answer: C

Rationale: The lesson states that equilibrium occurs when forward and reverse rates intersect.

MC 69. What does the lesson indicate about maintaining optimal conditions for enzymatic activity?

- A. It requires strong acids
- B. It depends on buffer mechanisms for pH control
- C. Temperature is the only factor
- D. It's unrelated to equilibrium

Correct answer: B

Rationale: The cause-effect summary links buffer mechanisms and pH control to maintenance of optimal conditions for enzymatic activity.

MC 70. According to the lesson, what characterizes the forward reaction rate decrease in ammonia synthesis?

- A. It's proportional to $[\text{NH}_3]^2$
- B. It's proportional to $[\text{N}_2][\text{H}_2]^3$
- C. It's independent of concentration
- D. It increases with time

Correct answer: B

Rationale: The lesson states that the forward rate decreases proportionally to $[\text{N}_2][\text{H}_2]^3$ as reactants are consumed.

MC 71. What does the lesson say about Henderson-Hasselbalch calculations?

- A. They only apply to strong acids
- B. They are used for buffer system design and pH calculations
- C. They replace Le Chatelier's principle
- D. They only work at high temperatures

Correct answer: B

Rationale: The lesson mentions Henderson-Hasselbalch calculations in the context of designing buffer systems for pH control.

MC 72. According to the lesson, what enables chemical engineers to optimize the Haber-Bosch process?

- A. Ignoring equilibrium
- B. Understanding equilibrium principles
- C. Using only high temperatures
- D. Avoiding catalysts

Correct answer: B

Rationale: The lesson states that understanding equilibrium principles allows chemical engineers to optimize reaction conditions.

MC 73. What does the lesson indicate about buffer capacity considerations?

- A. They are irrelevant to pH control
- B. They are important for designing effective buffer systems
- C. They only apply to strong acids
- D. They depend on color changes

Correct answer: B

Rationale: The lesson mentions buffer capacity considerations as important for designing buffer systems, particularly in the artificial blood substitute example.

MC 74. According to the lesson, what determines whether a system needs Le Chatelier's principle versus buffer calculations?

- A. Temperature only
- B. The specific context and type of equilibrium system
- C. Container volume
- D. Time of day

Correct answer: B

Rationale: The Final Diploma Alert advises practicing to identify when Le Chatelier's principle applies versus when buffer calculations are needed.

MC 75. What does the lesson say about the relationship between equilibrium understanding and global food security?

- A. They are unrelated
- B. Equilibrium understanding enables efficient fertilizer production affecting food security
- C. Food security prevents equilibrium
- D. Only temperature matters for food production

Correct answer: B

Rationale: The lesson explicitly states that without equilibrium understanding, efficient fertilizer production would be impossible, directly impacting global food security.

Unit D Review
Constructed Response Bank

Question 1

A student claims, 'At chemical equilibrium, all molecular motion stops because the concentrations are constant.'

- a. Explain why this claim is incorrect, describing what actually occurs at the molecular level during equilibrium.
- b. Analyze how the dance floor analogy from the lesson illustrates the dynamic nature of equilibrium.
- c. State one observable property besides concentration that remains constant at equilibrium.

Question 2

In the Haber-Bosch process, an engineer observes that increasing pressure from 150 atm to 200 atm increases ammonia yield.

- a. Explain how Le Chatelier's principle accounts for the increased ammonia yield when pressure increases.
- b. Analyze what happens to the equilibrium constant K_c when pressure changes from 150 to 200 atm at constant temperature.
- c. State the temperature range used in the industrial Haber-Bosch process.

Question 3

A researcher states that 'The N_2 triple bond with energy 945 kJ/mol must be overcome to form ammonia with N-H bonds of 391 kJ/mol each.'

- a. Explain the role of bond energies in determining whether the ammonia equilibrium favors reactants or products.
- b. Analyze how temperature affects the fraction of molecules able to overcome the activation energy barrier.
- c. State the bond energy of H_2 mentioned in the ammonia synthesis.

Question 4

A chemistry student observes that blood pH must remain between 7.35-7.45 for proper physiological function.

- a. Explain how buffer systems maintain this narrow pH range when acids are produced during metabolism.
- b. Analyze why both weak acid and conjugate base structures must be present for effective buffering.
- c. State one factor mentioned that affects buffer effectiveness.

Question 5

An industrial chemist notes that ammonia synthesis reaches equilibrium after several minutes to hours depending on conditions.

- a. Describe the changes in forward and reverse reaction rates as the ammonia system approaches equilibrium from pure reactants.
- b. Explain how collision frequency relates to the decreasing forward reaction rate.
- c. State one factor mentioned that affects how long it takes to reach equilibrium.

Question 6

A student claims that 'Equilibrium constants change whenever we add more reactants to shift the equilibrium position.'

- a. Explain why this claim is incorrect, stating what actually determines equilibrium constant values.
- b. Analyze the difference between equilibrium position and equilibrium constant when reactants are added.
- c. State what type of change can alter both equilibrium position and equilibrium constant.

Question 7

Research shows that the Haber-Bosch process operates at 150-200 atm and 400-500°C to achieve economically viable ammonia production.

- a. Explain why understanding equilibrium principles is essential for economically viable fertilizer production.
- b. Analyze the balance required between achieving high equilibrium concentrations and maintaining reasonable reaction rates.
- c. State the pressure range used in the industrial Haber-Bosch process.

Question 8

A researcher observes that at equilibrium, reactant concentrations decrease exponentially while product concentration increases logarithmically.

- a. Describe what happens to these concentration curves when the system reaches equilibrium.
- b. Explain why the forward rate decreases while the reverse rate increases during equilibrium establishment.
- c. State what mathematical relationship describes the forward rate's dependence on concentration.

Question 9

A student says, 'Closed systems are required for equilibrium because otherwise the reaction would stop.'

- a. Explain the actual reason why closed systems are essential for maintaining chemical equilibrium.
- b. Analyze what would happen to equilibrium if products continuously escaped from the system.
- c. State one observable property that must remain constant for equilibrium to exist.

Question 10

In Brønsted-Lowry acid-base theory, a chemist notes that reaction direction depends on relative proton affinities.

- a. Explain what determines the direction of proton transfer in acid-base reactions.
- b. Analyze how molecular structure creates the ability for proton donation or acceptance.
- c. State what type of reaction characterizes Brønsted-Lowry acid-base behavior.

Question 11

A biochemist observes that living systems maintain precise chemical equilibrium through dynamic balance of opposing processes.

- a. Explain how cells control equilibrium positions to maintain proper function.
- b. Analyze why dynamic balance rather than static conditions is essential for cellular function.
- c. State one mechanism cells use to resist pH disruption.

Question 12

Research indicates that thermodynamic principles govern equilibrium, with systems evolving toward minimum free energy while maximizing entropy.

- a. Explain how the balance between enthalpy and entropy determines equilibrium position.

- b. Analyze why systems naturally evolve toward states of minimum free energy.
- c. State what happens to entropy as systems approach equilibrium.

Question 13

An engineer observes that without understanding equilibrium, efficient large-scale fertilizer production would be impossible.

- a. Explain the connection between equilibrium understanding and global food security.
- b. Analyze how chemical engineers use equilibrium principles to optimize ammonia synthesis conditions.
- c. State the industrial process used for large-scale ammonia synthesis.

Question 14

A student observes that adding NH_3 to the equilibrium $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ causes a shift.

- a. Explain the immediate effect on forward and reverse reaction rates when NH_3 is added.
- b. Analyze how Le Chatelier's principle explains the system's response to added product.
- c. State what happens to the equilibrium constant K_c when NH_3 is added at constant temperature.

Question 15

Research shows that higher temperatures increase both forward and reverse reaction rates in equilibrium systems.

- a. Explain how temperature affects molecular collisions and reaction rates using the Maxwell-Boltzmann distribution.
- b. Analyze why equilibrium constants depend only on temperature and not on concentration or pressure.
- c. State what distribution describes molecular kinetic energies in the system.

Question 16

A researcher designing artificial blood notes that pH must be controlled through buffer systems incorporating Henderson-Hasselbalch calculations.

- a. Explain how a three-component buffer system would respond to metabolic acid production.
- b. Analyze how buffer capacity considerations affect the design of artificial blood substitutes.
- c. State the pH range that blood must maintain for proper function.

Question 17

A student claims that 'Equilibrium means the forward reaction has stopped because concentrations are no longer changing.'

- a. Explain why concentrations remain constant at equilibrium despite ongoing reactions.
- b. Analyze the relationship between reaction rates and observable properties at equilibrium.
- c. State what type of balance characterizes chemical equilibrium.

Question 18

Industrial data shows ammonia formation from N_2 and H_2 requires breaking a 945 kJ/mol triple bond.

- a. Describe the energy requirements for converting N_2 and H_2 into ammonia.
- b. Analyze how activation energy and collision frequency together determine reaction rates.
- c. State the energy of each N-H bond formed in ammonia.

Question 19

A biochemist observes that enzymatic activity requires precise pH control through buffer mechanisms.

- Explain how buffer systems enable optimal conditions for enzymatic activity.
- Analyze the connection between molecular structure and buffer effectiveness.
- State what mathematical relationship is used for buffer pH calculations.

Question 20

A chemistry professor states that 'Equilibrium systems achieve balance through rate equality, not through stopping reactions.'

- Explain what occurs at the molecular level when forward and reverse rates become equal.
- Analyze how this rate equality creates the characteristic features of chemical equilibrium.
- State what must be true about the system for equilibrium to be maintained.

Model Answers & Rubrics

Question 1

Part a: Explain why this claim is incorrect, describing what actually occurs at the molecular level during equilibrium.

Model Answer: The claim is incorrect because chemical equilibrium is a dynamic state, not static. At equilibrium, forward and reverse reactions continue occurring at equal rates. Individual molecules continuously react - reactants form products while products decompose back to reactants simultaneously. The concentrations remain constant because the rate of formation equals the rate of decomposition, creating a dynamic balance where molecular motion and reactions persist. (2 marks) (2 marks)

Part b: Analyze how the dance floor analogy from the lesson illustrates the dynamic nature of equilibrium.

Model Answer: The dance floor analogy demonstrates dynamic equilibrium by showing people entering and leaving at exactly the same rate. While the total number of dancers remains constant (like concentration at equilibrium), individual people are continuously moving in and out (like molecules reacting). This illustrates that equilibrium maintains constant observable properties through balanced opposing processes, not through cessation of activity. The continuous movement of individuals while maintaining overall constancy perfectly captures the dynamic nature. (2 marks) (2 marks)

Part c: State one observable property besides concentration that remains constant at equilibrium.

Model Answer: Temperature (or color, or pressure) remains constant at equilibrium. (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for identifying equilibrium as dynamic not static, 1 mark for explaining equal forward/reverse rates with ongoing reactions

Part b (2 marks): 1 mark for explaining constant numbers with individual movement, 1 mark for connecting to molecular behavior

Part c (1 mark): 1 mark for correctly identifying temperature, color, or pressure

Question 2

Part a: Explain how Le Chatelier's principle accounts for the increased ammonia yield when pressure increases.

Model Answer: According to Le Chatelier's principle, when pressure increases, the system shifts to counteract this change by favoring the side with fewer gas molecules. In $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$, there are 4 moles of gas on the left and 2 moles on the right. The system shifts right (toward products) to reduce pressure by decreasing the total number of gas molecules, increasing ammonia yield. (2 marks) (2 marks)

Part b: Analyze what happens to the equilibrium constant K_c when pressure changes from 150 to 200 atm at constant temperature.

Model Answer: The equilibrium constant K_c remains unchanged when pressure changes at constant temperature. K_c depends only on temperature, not on pressure or concentration changes. While the equilibrium position shifts to favor products (changing the actual concentrations), the ratio of products to reactants at the new equilibrium position still equals the same K_c value. The system adjusts concentrations to maintain the constant K_c . (2 marks) (2 marks)

Part c: State the temperature range used in the industrial Haber-Bosch process.

Model Answer: 400-500°C (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for identifying shift toward fewer moles of gas, 1 mark for explaining the shift right increases ammonia

Part b (2 marks): 1 mark for stating K_c unchanged, 1 mark for explaining K_c depends only on temperature

Part c (1 mark): 1 mark for stating 400-500°C

Question 3

Part a: Explain the role of bond energies in determining whether the ammonia equilibrium favors reactants or products.

Model Answer: Bond energies determine the overall energy balance of the reaction. Energy must be supplied to break the strong N_2 triple bond (945 kJ/mol) and H_2 bonds (436 kJ/mol), while energy is released forming N-H bonds (391 kJ/mol each). The overall energy balance between bond breaking (endothermic) and bond forming (exothermic) determines whether the equilibrium favors reactants or products based on thermodynamic favorability. (2 marks) (2 marks)

Part b: Analyze how temperature affects the fraction of molecules able to overcome the activation energy barrier.

Model Answer: Temperature affects equilibrium through the Maxwell-Boltzmann distribution of molecular kinetic energies. Higher temperatures increase the fraction of molecules possessing sufficient energy to overcome activation barriers for both forward and reverse reactions. This shifts the distribution curve, giving more molecules the minimum energy needed for successful collisions, thereby affecting both reaction rates and ultimately the equilibrium position. (2 marks) (2 marks)

Part c: State the bond energy of H_2 mentioned in the ammonia synthesis.

Model Answer: 436 kJ/mol (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for explaining energy input for bond breaking vs output for forming, 1 mark for connecting to equilibrium favorability

Part b (2 marks): 1 mark for mentioning Maxwell-Boltzmann distribution, 1 mark for explaining increased fraction with sufficient energy

Part c (1 mark): 1 mark for stating 436 kJ/mol

Question 4

Part a: Explain how buffer systems maintain this narrow pH range when acids are produced during metabolism.

Model Answer: Buffer systems maintain pH through weak acid-conjugate base equilibria that chemically neutralize added acids. When metabolic processes produce acids, the conjugate base component of the buffer accepts protons, converting to the weak acid form. This neutralization prevents significant pH change by removing free H^+ ions from solution, maintaining the delicate pH balance required for physiological function. (2 marks) (2 marks)

Part b: Analyze why both weak acid and conjugate base structures must be present for effective buffering.

Model Answer: Both components are essential because they handle opposite pH changes. The weak acid component neutralizes added bases by donating protons, while the conjugate base neutralizes added acids by accepting protons. Without both present, the system could only resist changes in one direction. The equilibrium between these two forms allows bidirectional pH protection through reversible proton transfer reactions. (2 marks) (2 marks)

Part c: State one factor mentioned that affects buffer effectiveness.

Model Answer: Having both weak acid and conjugate base structures available (or buffer capacity considerations) (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for explaining conjugate base accepts protons, 1 mark for describing neutralization preventing pH change

Part b (2 marks): 1 mark for explaining each component's role, 1 mark for explaining bidirectional protection

Part c (1 mark): 1 mark for stating appropriate factor from lesson

Question 5

Part a: Describe the changes in forward and reverse reaction rates as the ammonia system approaches equilibrium from pure reactants.

Model Answer: Initially, the forward rate is high due to maximum reactant concentrations while the reverse rate is zero (no products exist). As the reaction proceeds, the forward rate decreases proportionally to $[N_2][H_2]^3$ as reactants are consumed. Simultaneously, the reverse rate increases proportionally to $[NH_3]^2$ as ammonia accumulates. Equilibrium is reached when these rates become equal. (2 marks) (2 marks)

Part b: Explain how collision frequency relates to the decreasing forward reaction rate.

Model Answer: As $[N_2]$ and $[H_2]$ decrease during the reaction, there are fewer reactant molecules per unit volume. This reduces the frequency of collisions between N_2 and H_2 molecules. Since reaction rate depends on collision frequency (more collisions = more reactions), the decreasing collision frequency directly causes the forward reaction rate to decrease as equilibrium is approached. (2 marks) (2 marks)

Part c: State one factor mentioned that affects how long it takes to reach equilibrium.

Model Answer: Temperature (or catalyst presence) (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for describing initial rates, 1 mark for explaining changes until equal

Part b (2 marks): 1 mark for connecting concentration to collision frequency, 1 mark for linking to reaction rate

Part c (1 mark): 1 mark for stating temperature or catalyst

Question 6

Part a: Explain why this claim is incorrect, stating what actually determines equilibrium constant values.

Model Answer: The claim is incorrect because equilibrium constants (K_c) depend only on temperature, not on concentration changes. Adding reactants shifts the equilibrium position (actual concentrations change) but does not change the K_c value. The system adjusts concentrations to maintain the same K_c ratio. Only temperature changes can alter the equilibrium constant value. (2 marks) (2 marks)

Part b: Analyze the difference between equilibrium position and equilibrium constant when reactants are added.

Model Answer: Equilibrium position refers to the actual concentrations of reactants and products at equilibrium, which shift when reactants are added (system shifts right to form more products). The equilibrium constant is the mathematical ratio that these concentrations must satisfy, remaining unchanged. The position changes to maintain the constant K_c value - concentrations adjust but their ratio at the new equilibrium equals the same K_c . (2 marks) (2 marks)

Part c: State what type of change can alter both equilibrium position and equilibrium constant.

Model Answer: Temperature change (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for stating K_c depends only on temperature, 1 mark for explaining position shifts but K_c unchanged

Part b (2 marks): 1 mark for distinguishing position from constant, 1 mark for explaining maintenance of ratio

Part c (1 mark): 1 mark for identifying temperature

Question 7

Part a: Explain why understanding equilibrium principles is essential for economically viable fertilizer production.

Model Answer: Understanding equilibrium allows chemical engineers to optimize reaction conditions for maximum ammonia yield while maintaining reasonable reaction rates. Without this knowledge, efficient large-scale fertilizer production would be impossible. Engineers use equilibrium principles to balance yield, rate, and energy costs, directly impacting global food security through affordable fertilizer availability. (2 marks) (2 marks)

Part b: Analyze the balance required between achieving high equilibrium concentrations and maintaining reasonable reaction rates.

Model Answer: Higher temperatures increase reaction rates (more molecules overcome activation barriers) but may decrease equilibrium yield for exothermic reactions. Higher pressures favor ammonia formation (fewer gas molecules) but require expensive equipment. The 400-500°C and 150-200 atm conditions represent an economic optimization - sufficient rate for practical production while achieving acceptable equilibrium yields without excessive energy or equipment costs. (2 marks) (2 marks)

Part c: State the pressure range used in the industrial Haber-Bosch process.

Model Answer: 150-200 atm (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for optimization of conditions, 1 mark for connecting to food security

Part b (2 marks): 1 mark for temperature-rate vs yield tradeoff, 1 mark for pressure considerations

Part c (1 mark): 1 mark for stating 150-200 atm

Question 8

Part a: Describe what happens to these concentration curves when the system reaches equilibrium.

Model Answer: When equilibrium is reached, both concentration curves level off at constant values. The exponential decrease of reactants and logarithmic increase of products both flatten to horizontal lines. This indicates that forward and reverse reaction rates have become equal, with no further net change in concentrations despite ongoing molecular reactions. The curves remain flat as long as conditions are unchanged. (2 marks) (2 marks)

Part b: Explain why the forward rate decreases while the reverse rate increases during equilibrium establishment.

Model Answer: The forward rate decreases because reactant concentrations ($[N_2]$ and $[H_2]$) decrease, reducing collision frequency between reactant molecules. The forward rate is proportional to $[N_2][H_2]^3$. Simultaneously, the reverse rate increases because product concentration ($[NH_3]$) increases, providing more ammonia molecules capable of decomposing. The reverse rate is proportional to $[NH_3]^2$. These opposite trends continue until rates equalize. (2 marks) (2 marks)

Part c: State what mathematical relationship describes the forward rate's dependence on concentration.

Model Answer: Proportional to $[N_2][H_2]^3$ (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for describing leveling off, 1 mark for explaining constant values at equilibrium

Part b (2 marks): 1 mark for explaining forward rate decrease, 1 mark for explaining reverse rate increase

Part c (1 mark): 1 mark for stating $[N_2][H_2]^3$

Question 9

Part a: Explain the actual reason why closed systems are essential for maintaining chemical equilibrium.

Model Answer: Closed systems are essential because they prevent reactants or products from escaping or new substances from entering, which would disrupt the delicate balance of forward and reverse processes. If products escape, the reverse reaction rate decreases, preventing equilibrium. If reactants escape, the forward rate decreases. The closed system maintains constant amounts of all species, allowing rates to equalize. (2 marks) (2 marks)

Part b: Analyze what would happen to equilibrium if products continuously escaped from the system.

Model Answer: If products escaped, the product concentration would decrease, reducing the reverse reaction rate (proportional to product concentration). The forward reaction would continue converting reactants to products, but without sufficient products for the reverse reaction, rates could never equalize. The system would shift continuously toward products trying to reestablish equilibrium but never achieve it, eventually converting all reactants if given enough time. (2 marks) (2 marks)

Part c: State one observable property that must remain constant for equilibrium to exist.

Model Answer: Concentration (or color, temperature, or pressure) (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for preventing escape/entry, 1 mark for explaining balance disruption

Part b (2 marks): 1 mark for reduced reverse rate, 1 mark for inability to reach equilibrium

Part c (1 mark): 1 mark for valid property

Question 10

Part a: Explain what determines the direction of proton transfer in acid-base reactions.

Model Answer: The direction of proton transfer is determined by relative proton affinities and equilibrium constants. Protons transfer from species with lower proton affinity (stronger acids) to species with higher proton affinity (stronger bases). The equilibrium position favors the formation of weaker acids and weaker bases, with the equilibrium constant quantifying the extent of proton transfer. (2 marks) (2 marks)

Part b: Analyze how molecular structure creates the ability for proton donation or acceptance.

Model Answer: Molecular structure determines acid-base behavior through electronegativity patterns that create polar bonds. Highly electronegative atoms create partial charges that either stabilize negative charge after proton loss (acids) or provide electron density for proton acceptance (bases). These structural features determine whether a molecule can donate or accept protons and how readily it does so. (2 marks) (2 marks)

Part c: State what type of reaction characterizes Brønsted-Lowry acid-base behavior.

Model Answer: Proton transfer between molecules (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for relative affinities, 1 mark for favoring weaker acid/base formation

Part b (2 marks): 1 mark for electronegativity patterns, 1 mark for explaining stabilization/electron density

Part c (1 mark): 1 mark for proton transfer

Question 11

Part a: Explain how cells control equilibrium positions to maintain proper function.

Model Answer: Cells control equilibrium positions through concentration changes, temperature regulation, and buffer systems that resist pH disruption. By adjusting reactant and product concentrations through synthesis or degradation, cells shift equilibria. Temperature regulation affects reaction rates and equilibrium constants. Buffer systems maintain optimal pH for enzyme function, ensuring biochemical equilibria proceed correctly. (2 marks) (2 marks)

Part b: Analyze why dynamic balance rather than static conditions is essential for cellular function.

Model Answer: Dynamic balance allows cells to respond to changing conditions while maintaining homeostasis. Static conditions would prevent adaptation to metabolic demands. Dynamic equilibrium enables continuous nutrient processing, waste removal, and energy production while maintaining constant overall conditions. The ongoing molecular activity allows rapid responses to stimuli through equilibrium shifts while preserving vital concentration ratios. (2 marks) (2 marks)

Part c: State one mechanism cells use to resist pH disruption.

Model Answer: Buffer systems (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for listing control mechanisms, 1 mark for explaining their effects

Part b (2 marks): 1 mark for adaptability, 1 mark for explaining continuous processes

Part c (1 mark): 1 mark for buffer systems

Question 12

Part a: Explain how the balance between enthalpy and entropy determines equilibrium position.

Model Answer: Equilibrium position is determined by the balance between enthalpy changes (bond breaking and forming energy) and entropy changes (molecular disorder). Systems minimize free energy ($G = H - TS$) by balancing energy release/absorption with disorder changes. Favorable enthalpy changes (exothermic) and increased entropy both drive reactions forward, while their balance at given temperature determines final equilibrium position. (2 marks) (2 marks)

Part b: Analyze why systems naturally evolve toward states of minimum free energy.

Model Answer: Systems evolve toward minimum free energy because this represents the most thermodynamically stable state. At minimum free energy, the system has optimized the balance between energy content and entropy, requiring no net energy flow. This state represents maximum probability in molecular arrangements, where forward and reverse processes balance perfectly, creating the dynamic equilibrium observed in chemical systems. (2 marks) (2 marks)

Part c: State what happens to entropy as systems approach equilibrium.

Model Answer: Systems maximize entropy while minimizing free energy (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for explaining enthalpy/entropy components, 1 mark for balance determining position

Part b (2 marks): 1 mark for stability concept, 1 mark for probability/balance explanation

Part c (1 mark): 1 mark for maximizing entropy

Question 13

Part a: Explain the connection between equilibrium understanding and global food security.

Model Answer: Equilibrium understanding enables optimization of the Haber-Bosch process for ammonia synthesis, the foundation of nitrogen fertilizer production. Efficient fertilizer production makes agriculture productive enough to feed global populations. Without equilibrium principles to maximize yield and minimize costs, fertilizer would be too expensive or scarce, leading to reduced crop yields and food insecurity worldwide. (2 marks) (2 marks)

Part b: Analyze how chemical engineers use equilibrium principles to optimize ammonia synthesis conditions.

Model Answer: Engineers apply Le Chatelier's principle to choose high pressure (150-200 atm) favoring product formation (fewer gas molecules). They balance temperature (400-500°C) to achieve reasonable reaction rates while maintaining acceptable equilibrium yield. They calculate equilibrium constants to predict yields under different conditions. This optimization maximizes ammonia production while minimizing energy costs and equipment requirements. (2 marks) (2 marks)

Part c: State the industrial process used for large-scale ammonia synthesis.

Model Answer: The Haber-Bosch process (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for fertilizer-food connection, 1 mark for economic/availability impact

Part b (2 marks): 1 mark for pressure/temperature optimization, 1 mark for yield/cost balance

Part c (1 mark): 1 mark for Haber-Bosch process

Question 14

Part a: Explain the immediate effect on forward and reverse reaction rates when NH_3 is added.

Model Answer: Adding NH_3 immediately increases the reverse reaction rate because it's proportional to $[\text{NH}_3]^2$. The forward rate remains initially unchanged as reactant concentrations haven't changed yet. This rate imbalance causes net reverse reaction (NH_3 decomposition to N_2 and H_2) until rates equalize again at a new equilibrium position with higher reactant concentrations. (2 marks) (2 marks)

Part b: Analyze how Le Chatelier's principle explains the system's response to added product.

Model Answer: Le Chatelier's principle states the system shifts to counteract the imposed change. Adding product (NH_3) disturbs equilibrium, so the system shifts left (reverse direction) to consume excess NH_3 . This reduces product concentration while increasing reactant concentrations. The shift continues until a new equilibrium position is established where the rate equality is restored at different concentrations. (2 marks) (2 marks)

Part c: State what happens to the equilibrium constant K_c when NH_3 is added at constant temperature.

Model Answer: K_c remains unchanged (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for increased reverse rate, 1 mark for rate imbalance explanation

Part b (2 marks): 1 mark for counteracting change, 1 mark for shift left explanation

Part c (1 mark): 1 mark for K_c unchanged

Question 15

Part a: Explain how temperature affects molecular collisions and reaction rates using the Maxwell-Boltzmann distribution.

Model Answer: Higher temperatures shift the Maxwell-Boltzmann distribution of molecular kinetic energies to higher values. This increases the fraction of molecules possessing sufficient energy to overcome activation barriers for both forward and reverse reactions. More molecules have the minimum energy for successful collisions, increasing collision effectiveness and thus both reaction rates, though not necessarily equally. (2 marks) (2 marks)

Part b: Analyze why equilibrium constants depend only on temperature and not on concentration or pressure.

Model Answer: Equilibrium constants reflect the thermodynamic favorability determined by free energy differences between reactants and products. This energy difference depends on molecular structures and temperature (affecting enthalpy and entropy contributions) but not on concentrations or pressure. While concentration and pressure changes shift equilibrium position, the fundamental energy relationships that K_c represents remain constant at fixed temperature. (2 marks) (2 marks)

Part c: State what distribution describes molecular kinetic energies in the system.

Model Answer: Maxwell-Boltzmann distribution (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for distribution shift, 1 mark for increased fraction above activation energy

Part b (2 marks): 1 mark for thermodynamic basis, 1 mark for independence from concentration/pressure

Part c (1 mark): 1 mark for Maxwell-Boltzmann

Question 16

Part a: Explain how a three-component buffer system would respond to metabolic acid production.

Model Answer: When metabolic acids are produced, the conjugate base components of the buffer system accept protons, converting to their weak acid forms. This neutralization removes free H^+ from solution, preventing pH decrease. Multiple buffer components provide broader pH protection and greater capacity. Each weak acid-conjugate base pair operates at its optimal pH range, collectively maintaining blood pH at 7.35-7.45. (2 marks) (2 marks)

Part b: Analyze how buffer capacity considerations affect the design of artificial blood substitutes.

Model Answer: Buffer capacity determines how much acid or base the system can neutralize before significant pH change occurs. Higher capacity requires optimal concentrations of both weak acid and conjugate base forms. For artificial blood, capacity must handle normal metabolic acid production and respiratory CO_2 changes. Design must balance sufficient capacity with physiological compatibility and osmotic considerations. (2 marks) (2 marks)

Part c: State the pH range that blood must maintain for proper function.

Model Answer: 7.35-7.45 (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for conjugate base accepting protons, 1 mark for multiple components providing protection

Part b (2 marks): 1 mark for capacity definition, 1 mark for design considerations

Part c (1 mark): 1 mark for 7.35-7.45

Question 17

Part a: Explain why concentrations remain constant at equilibrium despite ongoing reactions.

Model Answer: Concentrations remain constant because forward and reverse reactions occur at exactly equal rates, not because reactions have stopped. For every molecule of reactant converting to product, a molecule of product simultaneously converts back to reactant. This creates dynamic balance where the net change is zero, maintaining constant concentrations despite continuous molecular activity and reactions in both directions. (2 marks) (2 marks)

Part b: Analyze the relationship between reaction rates and observable properties at equilibrium.

Model Answer: When forward and reverse rates are equal, all observable properties (concentration, color, temperature, pressure) remain constant over time. The rate equality means no net production or consumption of any species, so bulk properties don't change. However, this constancy masks the underlying dynamic process where individual molecules continuously react, demonstrating that macroscopic stability results from microscopic balance. (2 marks) (2 marks)

Part c: State what type of balance characterizes chemical equilibrium.

Model Answer: Dynamic balance (not static balance) (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for equal rates not stopped, 1 mark for explaining net zero change

Part b (2 marks): 1 mark for constant properties from rate equality, 1 mark for dynamic vs observable

Part c (1 mark): 1 mark for dynamic balance

Question 18

Part a: Describe the energy requirements for converting N₂ and H₂ into ammonia.

Model Answer: Energy must be supplied to break the N≡N triple bond (945 kJ/mol) and H-H bonds (436 kJ/mol each). This energy input is partially offset by energy released when forming six N-H bonds (391 kJ/mol each) in two NH₃ molecules. The overall energy balance between bond breaking (endothermic) and bond forming (exothermic) determines the reaction's thermodynamic favorability. (2 marks) (2 marks)

Part b: Analyze how activation energy and collision frequency together determine reaction rates.

Model Answer: Reaction rate depends on both having sufficient collisions (frequency) and those collisions being energetic enough (activation energy). Higher concentration increases collision frequency, providing more opportunities for reaction. However, only collisions where molecules possess energy exceeding the activation barrier lead to reaction. Temperature affects both by increasing molecular speeds (frequency) and the fraction exceeding activation energy. (2 marks) (2 marks)

Part c: State the energy of each N-H bond formed in ammonia.

Model Answer: 391 kJ/mol (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for bond breaking requirements, 1 mark for energy balance concept

Part b (2 marks): 1 mark for frequency and energy requirements, 1 mark for temperature effects

Part c (1 mark): 1 mark for 391 kJ/mol

Question 19

Part a: Explain how buffer systems enable optimal conditions for enzymatic activity.

Model Answer: Buffer systems maintain constant pH through weak acid-conjugate base equilibria that neutralize acids and bases produced during metabolism. Enzymes have optimal pH ranges where their active sites maintain proper shape and charge distribution. Buffers prevent pH fluctuations that would

denature enzymes or alter their catalytic efficiency, ensuring biochemical reactions proceed at appropriate rates for cellular function. (2 marks) (2 marks)

Part b: Analyze the connection between molecular structure and buffer effectiveness.

Model Answer: Buffer effectiveness depends on having both weak acid and conjugate base structures with appropriate pKa values near the target pH. Molecular structure determines acid strength through electronegativity patterns and resonance stabilization. The structural features that create moderate proton affinity enable reversible proton transfer, allowing the buffer to respond to pH changes in either direction effectively. (2 marks) (2 marks)

Part c: State what mathematical relationship is used for buffer pH calculations.

Model Answer: Henderson-Hasselbalch equation (or relationships between pH, pOH, Ka, Kb, and Kw) (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for pH maintenance mechanism, 1 mark for enzyme function connection

Part b (2 marks): 1 mark for structural requirements, 1 mark for reversible proton transfer

Part c (1 mark): 1 mark for Henderson-Hasselbalch or pH relationships

Question 20

Part a: Explain what occurs at the molecular level when forward and reverse rates become equal.

Model Answer: When rates become equal, reactant molecules continue converting to products at the same rate that product molecules decompose back to reactants. Individual molecules remain in constant motion, colliding and reacting. The equal rates mean that for every forward reaction occurring, a reverse reaction simultaneously occurs elsewhere in the system, creating no net change in concentrations despite continuous molecular activity. (2 marks) (2 marks)

Part b: Analyze how this rate equality creates the characteristic features of chemical equilibrium.

Model Answer: Rate equality produces constant concentrations because production and consumption balance perfectly. This creates all observable equilibrium characteristics: constant properties (color, temperature, pressure), reproducible equilibrium constants, and predictable responses to disturbances. The dynamic nature allows the system to respond to changes through rate adjustments while the equality ensures stability under constant conditions. (2 marks) (2 marks)

Part c: State what must be true about the system for equilibrium to be maintained.

Model Answer: The system must be closed (1 mark) (1 marks)

Scoring rubric

Part a (2 marks): 1 mark for ongoing reactions, 1 mark for no net change explanation

Part b (2 marks): 1 mark for constant properties from balance, 1 mark for dynamic response capability

Part c (1 mark): 1 mark for closed system